Functional Stability Theory — A Programmatic Hub: Universal Convexity-Uniqueness Across Scales

From Particles to Cosmology

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May 2026

Abstract

Functional Stability Theory (FST) is a research programme organised around a single proposed stability principle – the *Universal Convexity-Uniqueness criterion* (UCU) – as a possible unifying framework for emergent structure across physical scales, from thermodynamic equilibria of fundamental parameters (FST-I) through autocatalytic chemistry (FST-II) and biological evolutionary stability (FST-III) to cosmological dark energy and turbulence. This overview synthesises the programme. The universal game (x_L, G_L, D_L) is summarised together with the Maximum Entropy Production Principle (MEPP) initial-condition axiom, and the three-lemma endgame UCU1 (strict convexity), UCU2 (existence of minimiser) and UCU3 (quantitative quadratic lower bound) is formulated as the technical core. The renormalisation group (RG) provides the mathematical bridge between scales. We tabulate the status of UCU1/UCU2/UCU3 across representative applications (thermodynamics, chemistry, biology, K41 turbulence, cosmological dark energy, Yang–Mills mass gap) and document the methodological lesson learned from the cosmological route (*Pattern A*): a positivity reformulation of an open conjecture usually amounts to a change of language rather than progress, unless its proof strength is independently justified. This article is an *honest survey*: it separates established structural contributions from conditional claims and records the open sub-problems (HD3' for Beal, full UCU3 for Yang–Mills, L4 RG-matching for the dark-energy/CRM-VI line).

Keywords: stability, entropy production, renormalisation group, Nash equilibrium, scale invariance, fine-tuning, dark energy, turbulence, mass gap.

1 Introduction

The fine-tuning problem – the empirical observation that a wide range of physical parameters (Standard-Model couplings, the cosmological constant, the chemical reaction networks compatible with life) appears to be selected with high precision among all logically possible values – is usually treated as scale-specific: anthropic at the cosmological scale, kinetic at the chemical scale, evolutionary at the biological scale. Functional Stability Theory (FST) explores the alternative hypothesis that these selection statements are different manifestations of a single stability principle.

The programme has a fixed shape across scales:

(i) A configuration space X on which a functional F is defined.

(ii) A conditional system: F admits a unique minimiser provided a strict convexity property (UCU1), an existence property (UCU2) and a quantitative quadratic lower bound (UCU3) hold on a closed convex feasibility set $K \subset X$.

(iii) A renormalisation-group (RG) bridge connecting two scales, intended to show when a minimiser at one scale induces a stability statement at another.

The contribution of this overview is therefore not a new theorem, but the *programmatic* proposal that scales as different as Standard-Model parameter selection (FST-I) and dark-energy late-time acceleration (CRM-VI) can be analysed through the same conditional structure. Earlier papers in the series treat individual scales; the present article maps them onto a single architecture and identifies which sub-problems remain open.

FST Series at a Glance

This paper serves as the *programmatic hub* for the Functional Stability Theory series. Table 1 lists the direct application-scale companion papers (FST-I, FST-II, FST-III, FST-DE) and the major published FST building blocks in the mathematics and physics lines that this overview references.

2 The Universal Game

2.1 Two Axioms

Axiom 1 (Initial Condition)**.** *The cosmos starts in a state of maximal gradient and minimal structure (low entropy, high free-energy gradient).*

Axiom 2 (Evolution: MEPP)**.** *Open non-equilibrium subsystems evolve, on the timescales of interest, along trajectories consistent with the Maximum Entropy Production Principle (MEPP) [1].*

2.2 Scale Triple

For each physical scale L we associate a triple

$$\mathcal{S}_L \;=\; (x_L,\; G_L,\; D_L),$$

where x_L is the configuration variable (field, population fraction, density profile), $G_L = (S_L, \Pi_L)$ is the local game (strategy space S_L and payoff Π_L), and D_L is the dynamics $\dot{x}_L = D_L(x_L; \Pi_L)$.

Paper	Scale / Target	Reference	Series role
Direct application companions (FST-PRACTICAL)			
FST-I	Thermodynamic / SM parameters	[8]	hasPart
FST-II	Chemical / autocatalysis	[9]	hasPart
FST-III	Biological / protein folding	[10]	hasPart
FST-IV	Cosmological collector slot	forthcoming	hasPart
$\hookrightarrow$ FST-DE	sub-paper: dark energy	[11]	sub-component
Mathematical building blocks (FST-MATHEMATICS)			
RH / Even Dominance	spectral stability	[12]	references
Zeta Zoo (FST mathematics)	classification	[13]	references
Zookeeper (CCM route)	RH technical core	[14]	references
FST Spectrum Duality	physics-master	[15]	references
Hodge Positivity	AP / algebraicity	[16]	references
BSD Positivity	elliptic curve heights	[17]	references
P vs NP Entropy	complexity	[18]	references
Physical building blocks (FST-PHYSICS)			
Yang–Mills Mass Gap	strong-coupling lattice	[19]	references
Navier–Stokes Regularity	3D NS attractor route	[20]	references
NS-LDI Framework	TLL/LDI bridge	[21]	references
K41 Turbulence (joint minimiser)	Kolmogorov spectrum	[22]	references
Turbulence / DFC cascade route	anomalous dissipation	[23]	references

Table 1: FST series components. "hasPart" (Zenodo Related Identifier `hasPart`) marks the four direct application-scale companions that are co-published with this overview. "references" marks published FST building blocks that the overview cites but does not formally bundle. All identifiers are persistent Zenodo DOIs; see the bibliography for resolvers.

2.3 Stability Definition

Definition 1 (Stable State)**.** *x_L^* is* stable *at scale L if it is a locally attractive fixed point of D_L (Lyapunov sense) and robust against invasion or perturbation (Nash/ESS criterion at the game level). This Nash/ESS language follows the standard evolutionary-game background of Maynard Smith [7].*

The choice of payoff proxy at each scale (entropy production $\dot{S}$, free-energy destruction $-\Delta G$, mean fitness $\bar{f}$, etc.) is dictated by the physical setting; the FST working hypothesis is that the *long-time stable* attractor correlates with the maximum payoff proxy under MEPP-compatible boundary conditions. The cosmological branch also uses the thermodynamic-spacetime intuition exemplified by Jacobson's equation-of-state reading of Einstein gravity [6].

3 The Universal Stability Criterion (UCU)

The technical core of the programme is a three-lemma endgame formulated in the convex-analysis language of reflexive Banach spaces. We refer to `CORE/UCU_FORMAL_DEFINITIONS.md` for the formal definitions and the standard literature (Brezis [2], Ekeland–Témam [3], Evans [4]). Throughout this section, X is a reflexive real Banach space, $K \subset X$ is non-empty, closed and convex, and $F : K \to \mathbb{R} \cup \{+\infty\}$ satisfies (A1) properness, (A2) weak lower semicontinuity, (A3) coercivity.

Lemma 1 (UCU1 – Strict Convexity)**.** *F is* strictly convex *on K: $F(\lambda x + (1-\lambda)y) < \lambda F(x) + (1-\lambda)F(y)$ for all $x \neq y$ in K and $\lambda \in (0, 1)$.*

Lemma 2 (UCU2 – Existence)**.** *Under (A1)–(A3), F attains its infimum on K: there exists $x^* \in K$ with $F(x^*) = \inf_K F$.*

Lemma 3 (UCU3 – Quantitative Quadratic Lower Bound)**.** *There exists $\mu > 0$ such that for all $x \in K$,*

$$F(x) \;\geq\; F(x^*) \;+\; \tfrac{\mu}{2}\,\|x - x^*\|^2.$$

UCU3 splits into a local *and a* global *version (see Section 5).*

Consequence (UCU-Theorem, conditional form). If UCU1 + UCU2 + UCU3 hold on K, then F admits a *unique* minimiser, and the minimum is *quantitatively isolated* in $\|\cdot\|$. The first two lemmas are classical (Tonelli direct method); UCU3 is the application-specific input that ties the abstract result to the physics of the scale.

Remark. The role of UCU3 is the same as the role of a *spectral gap* in the resolvent picture of [14]: it converts a qualitative uniqueness statement into a quantitative stability estimate. Across scales, UCU3 reads “no flat direction at the minimum”.

4 Renormalisation Group as Mathematical Bridge

The RG provides the formal apparatus for relating the universal game at scale L to that at scale $L{+}1$, in the Wilson–Kogut tradition of scale transformations [5]. A Kadanoff block transformation $R_b : (x_L, G_L, D_L) \mapsto (x_{L+1}, G_{L+1}, D_{L+1})$ is required to preserve at least one of the following invariants:

(A) the fixed-point structure (number and type of stable attractors),

(B) the best-response geometry (sign pattern of the Jacobian around x^*_L),

(C) the existence of a Lyapunov / potential function.

Whenever (A)–(C) is preserved, a UCU-stable scale-L minimiser induces a UCU-stable scale-($L{+}1$) minimiser.

Compact statement. The RG-bridge is the mechanism by which the conditional UCU theorem becomes *transferable*. In practice, the bridge has been worked out only in fragments (see the open problems in Section 7); for the cosmological RG-matching of the $R + \gamma R^2$ sector (CRM-VI), the L4 step remains open.

5 Cross-Scale Status Map

Table 2 summarises the present status of UCU1, UCU2, UCU3 (local and global) across the six application scales of the programme.

Reading the table. For thermodynamics, chemistry and biology, UCU1+UCU2 are obtained from convexity of free-energy-like functionals on physically motivated feasibility sets; UCU3 is local but not global, because non-equilibrium boundary conditions can produce competing basins. For K41 turbulence, the local UCU3 is achieved at the Kolmogorov scaling exponent $g^* = \frac{1}{2}$, but a logarithmic Sobolev inequality (LSI) is required to upgrade to a global statement. For dark energy (CRM-VI), UCU1–UCU3 hold for the de Sitter functional $\Phi(\rho)$ on $\mathbb{R}_+$, but lifting the result through the L4 RG-matching to a full effective field theory is not yet completed.

Scale	Functional	UCU1	UCU2	UCU3 local	UCU3 global
FST-I (Thermo / SM)	$\Sigma[\alpha, m_e/m_p, \dots]$	✓	✓	plateau	open
FST-II (Chemistry)	G[hypercycle]	✓	✓	local OK	open
FST-III (Biology / fold)	$F_{\text{fold}}[\phi]$	✓	✓	local OK	open
K41 (Turbulence)	$F[E(k)]$	✓	✓	✓$(g^* = \frac{1}{2})$	LSI open
DE / CRM-VI (Cosmology)	$\Phi(\rho)$	✓	✓	✓	L4 RG-match open
YM (Mass gap)	$\mathcal{E}[A]$	?	?	?	UCU3 = *mass-gap question*

Table 2: Status of UCU1/UCU2/UCU3 across application scales. ✓ = established conditional on standard inputs (Brezis, Ekeland–Témam direct method); "open" = sub-problem identified, not yet resolved within the FST framework. The Yang–Mills line is included to mark the analogue: UCU3 (global) for the YM functional *is* the mass-gap question itself.

5.1 Status of FST Building Blocks (mathematics and physics)

Beyond the UCU-application table above, the FST programme includes a set of published *building blocks* in the mathematics and physics lines. Table 3 summarises their honest proof status (drawn from each project's individual proof note, see the companion repository at `STATUS_UEBERSICHT.md`).

Building block	Line	Status	Open core gap
RH / Even Dominance	mathematics	unconditional	analytic c_{gap} derivation removes last numerical input
Zeta Zoo / FST mathematics	mathematics	conditional	Prime-Hub / OP5 obstructions
Hodge	mathematics	framework	Hard direction (AP ⇒ algebraic)
BSD	mathematics	framework	Higher Gross–Zagier for rank ≥ 2
P vs NP (Entropy)	mathematics	reformulation	Uniformity bridge (NOT a proof)
Yang–Mills mass gap	physics	strong-coupling proven	RG/continuum transport conditional
Navier–Stokes (Attractor)	physics	conditional framework	Attractor regularity package, Assumption G/G'
NS-LDI	physics	framework + proof of life	TLL+LDI for 3D NS attractors
K41 Turbulence (Paper A)	physics	unconditional	— (Paper B remains conditional on DFC1$^\vee$)
Turbulence / DFC cascade	physics	conditional framework	anomalous dissipation depends on DFC1$^\vee$ and DFC bridge closure
FST-DE (Dark Energy)	cosmology	framework + no-go	UV renormalisation, screening lift

Table 3: Honest proof status of FST building blocks in the mathematics, physics and cosmology lines, as of 2026-05. Every entry references an individual project whose primary proof note (BEWEISNOTIZ.md) documents the precise lemma chain and the residual open gaps. The present overview neither proves new instances of these results nor strengthens their mathematical content; it locates each within the same UCU/RG/MEPP architecture. Where the status reads "reformulation" (P vs NP) or "framework" (Hodge, BSD, NS, FST-DE), the open core gap is the original Millennium-class difficulty – consistent with the Pattern-A lesson (Section 6).

6 The Pattern-A Lesson

Across both the mathematics line (Hodge, Beal/abc, BSD, P vs NP) and the cosmology line (CRM-VI, dark energy), one methodological pattern has recurred often enough to require a programme-level *caution*.

> **Pattern A (negative form).** A reformulation of an open conjecture V into a "positivity normal form" P is, in general, a change of formal language rather than a proof strategy: typically $P \Leftrightarrow V$. Without an independent strength comparison, P should not be treated as a proof of V.

The cosmological variant is operationally identical: mapping a *physical* question V_{phys} onto a model M that contains an open mathematical sub-question S does not solve V_{phys}; it merely reduces it conditionally to S. The cosmological CC-problem and the inflationary initial-condition problem each met this pattern in the CRM-VI line. The lesson is recorded in `CORE/PATTERN_A_LESSON.md` and the companion internal note [25]; it is the reason why every UCU-application paper now contains a section labelled *Honest Status*.

7 Open Sub-Problems

The programme leaves three explicitly identified open problems.

HD3' for Beal. The Beal-Verifier-Sieve framework reduces the Beal conjecture to four sub-statements HD3'-A through HD3'-D. As documented in `HD3_B_EFFECTIVE_CHARACTER_SUMS.md` and `HD3_ACD_TRIO_OBSTRUCTIONS.md`, the strong character-sum bound HD3'-B (cancellation $q^{1+\varepsilon}$ rather than the standard square-root cancellation $q^{3/2}$) lies beyond standard Weil–Deligne tools, and HD3'-A (Salt-Closing) and HD3'-C (Counting) reduce to it. A note documenting the conditional structure of HD3'-A under HD3'-B is provided in `HD3_A_SALT_CLOSING.md`.

Global UCU3 for Yang–Mills. In the FST language, the mass-gap question for $4d$ pure Yang–Mills becomes the question of whether the energy functional $\mathcal{E}[A]$ admits a *global* quadratic lower bound around the vacuum: a global UCU3. This identification does not solve Yang–Mills; it locates the mass-gap question as the canonical UCU3-global representative. UCU1 and UCU2 in the YM context remain open within the FST language.

L4 RG-matching for CRM-VI. The cosmological dark-energy line (CRM-VI), developed in the CRM programme [24], supplies UCU1–UCU3 on a reduced configuration space (de Sitter functional in $\mathbb{R}_+$). Lifting the result through a four-step RG-match (L1 microscopic action, L2 effective potential, L3 cosmological background, L4 perturbations + matching to data) leaves L4 open. This is the cosmological analogue of the Pattern-A lesson: the mathematical reduction is conditional on a physical RG-match that has not yet been performed at the level required for a publishable physics claim.

8 Honest Status

What FST provides.

A common architecture (universal game + MEPP axiom + UCU three-lemma endgame + RG bridge) under which scales as different as the Standard-Model parameter space and dark-energy late-time acceleration admit a single conditional uniqueness/stability statement. A status map (Table 2) showing where the architecture is closed and where it is conditional. A methodological lesson (Pattern A) that separates structural reformulation from proof progress in the mathematics line.

What FST does *not* provide.

A solution to any individual hard open problem (Yang–Mills mass gap, Beal, BSD). FST fixes the *shape* of the conditional statement; the application-specific UCU3-global step is, in each hard case, identified with the original difficulty. The novelty is therefore architectural and reductive, not a constructive proof of those conjectures.

Falsification criterion.

The programme would face direct falsification pressure if, for any listed scale, UCU1 or UCU2 failed on the natural feasibility set. UCU3-global failure is not itself a falsification: it marks the open sub-problem at that scale.

Computational Note

This article is a programme overview. Computations supporting the cosmological line (Watkins-style spectrum experiments) were performed on `ellmos-services`; the present overview itself introduces no new numerical experiment.

AI Disclosure (Level L3)

This paper was prepared with substantial AI assistance (Anthropic Claude, OpenAI GPT, Google Gemini variants). AI tools were used for: cross-paper synthesis between the FST-I, FST-II, FST-III, FST-DE companions; structural editing and consistency checking of the UCU/RG architecture; generation and verification of the cross-scale status maps (Tables 1, 2 and 3); literature cross-referencing with Zenodo DOIs; and bilingual EN/DE synchronisation. All scientific claims, the UCU three-lemma architecture, the Pattern A lesson, and the falsification criterion are the author's own and are subject to standard authorial responsibility. No section was written autonomously by AI. The categorisation follows the FST programme's five-level AI disclosure scheme (L1 = no AI; L5 = AI as principal contributor); this paper is L3 (AI as substantial assistant, human as principal author and responsible scientist).

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